

Synaptic plasticity as short-term memory

A technical visual essay on fixed-state associative memory, interference, and the BDH connection.

The problem: memory grows with the sequence

Transformer decoding preserves prior keys and values explicitly. For long contexts, this produces a memory footprint that scales with sequence length and makes each new token increasingly dependent on memory traffic. A fixed recurrent state offers a different systems trade-off: context is compressed into the state rather than retained token-by-token.

Core question. Can a network store recent associations by changing its synaptic state, rather than allocating one new memory slot per token?

From attention to synaptic memory

A useful mathematical bridge is the outer-product accumulation used by linear attention / fast-weight formulations: a sequence of key-value pairs can be summarized by a matrix state. A Hebbian formulation writes an association with a local outer product:

$$W(t) = \lambda W(t-1) + \eta (V(t) \times K(t)^T)$$

Here λ controls decay and η controls plasticity. The attraction is constant state shape. The trade-off is that many associations now share the same representational substrate.

The failure mode: interference

For a cue Q_i , a dense memory retrieves the intended target plus contributions from other stored pairs. Those cross-terms are not a software bug; they are a mathematical consequence of superposing associations in one finite-dimensional state. When the number of effectively independent associations approaches the representation dimension, clean separation becomes harder.

Signal + cross-talk

The project artifact makes this visible with a small synthetic memory. The learner changes d and N and immediately sees recall fidelity move. This is the key educational substrate: the learner can reproduce the central claim instead of taking a static diagram on trust.

Why sparsity is relevant to BDH

The Pathway BDH paper describes working memory during inference as relying on synaptic plasticity with Hebbian learning, together with sparse and positive activation. In the conceptual picture used here, wider representations create more room for associations, while sparse activation limits how much of the synaptic graph is modified by each input. The purpose of the explainer is not to claim that sparsity magically removes all interference; it is to show why the architectural move is relevant.

Evidence, not marketing

The supplied brief asks for strong evidence discipline. Accordingly, BDH-specific benchmark claims are explicitly labeled as reported results. The official Pathway GitHub repository states that its public repository does not reproduce the reported 97.4% Sudoku Extreme result out of the box. That distinction matters: a developer-reported result is evidence, but it is not the same thing as independent reproduction.

Where the idea sits

The concept sits at the intersection of associative memory, fast weights, linear attention, recurrent state, and biologically inspired learning. Recent primary work also studies learned plasticity rules, memory addressing, and interacting excitatory/inhibitory plasticity, showing that plasticity remains an active research area rather than a purely historical analogy.

Takeaways

1. Synaptic state can compress sequence information into a fixed-shape recurrent memory.
2. The compression introduces a measurable capacity/interference trade-off.
3. BDH is relevant because it makes synaptic plasticity a first-class part of inference-time memory.
4. The honest engineering question is not whether the idea sounds brain-like, but whether capacity, interference, sparse updates and hardware cost can be balanced in practice.

Primary sources

Kosowski et al. (2025), arXiv:2509.26507 — The Dragon Hatchling.

Tyulmankov, Yang & Abbott (2022), Neuron 110(3):544-557.e8 — meta-learned synaptic plasticity and memory addressing.

Shervani-Tabar & Rosenbaum (2023), Nature Communications 14:1805 — biologically plausible learned plasticity rules.

Agnes & Vogels (2024), Nature Neuroscience 27:964-974 — co-dependent excitatory/inhibitory plasticity and memory stability.